ARIEL GOODWIN

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EDUCATION

Ph.D. Applied Mathematics 2022 – 2027 (Expected)
Cornell University, Center for Applied Mathematics GPA: 4.00/4.00

Committee: Adrian Lewis (Advisor), Anil Damle, and Soroosh Shafiee

M.Sc. Applied Mathematics 2022 – 2025
Cornell University, Center for Applied Mathematics GPA: 4.00/4.00

B.Sc. Honours Mathematics and Computer Science 2018 – 2022
McGill University, Department of Mathematics and Statistics GPA: 3.98/4.00

AWARDS & SCHOLARSHIPS

INFORMS Optimization Society Student Paper Prize 2026
Institute for Operations Research and the Management Sciences (INFORMS)

· Awarded for *Stochastic and incremental subgradient methods for convex optimization on Hadamard spaces*; recognizes an outstanding student paper in optimization published in the last three years

NSERC Postgraduate Scholarship - Doctoral (PGS D) 2024 – 2027
Natural Sciences and Engineering Research Council of Canada (NSERC), held at Cornell University

· To support doctoral research

Cornell Fellowship 2022
Cornell University

· Recruitment fellowship for academic merit, covering one year of Ph.D. studies

Sir Edward Beatty Memorial Scholarship in Mathematics 2021
McGill University

· For academic achievement in mathematics

NSERC Undergraduate Student Research Award Summer 2021
NSERC, held at McGill University

· Conducted research on convex optimization for statistical estimation, supervised by Rustum Choksi

NSERC Undergraduate Student Research Award Summer 2020
NSERC, held at McGill University

· Conducted research on nonsmooth optimization for projection problems, supervised by Tim Hoheisel

Dr. Feng Qian Scholarship 2020
McGill University

· For academic achievement in computer science

Seba Abbott Smith Scholarship 2018 – 2022
McGill University

· For academic merit and leadership in secondary and postsecondary school

RESEARCH VISITS

Visiting Graduate Student Summer 2026
Halıcıoğlu Data Science Institute, University of California San Diego

· Hosted by Dmitriy Drusvyatskiy for a research project on the geometry of decision trees

NSF Mathematical Sciences Graduate Internship Summer 2023
National Science Foundation (NSF), held at the National Laboratory of the Rockies (formerly NREL)

· Hosted by Jonathan Maack for a research project on manifold optimization for optimal power flow

REFEREED JOURNAL PUBLICATIONS (APPEARED OR ACCEPTED)

Circumcenters and mean sets in Hadamard space: horospherical subgradient methods
With A. S. Lewis, G. López-Acedo, and A. Nicolae. *Mathematics of Operations Research*, 2026.

Stochastic and incremental subgradient methods for convex optimization on Hadamard spaces
With A. S. Lewis, G. López-Acedo, and A. Nicolae. *Mathematical Programming*, 2026. Recipient of the **INFORMS Optimization Society's 2026 Student Paper Prize.**

Recognizing weighted means in geodesic spaces
With A. S. Lewis, G. López-Acedo, and A. Nicolae. *Foundations of Computational Mathematics*, 2025.

Convex optimization on CAT(0) cubical complexes
With A. S. Lewis, G. López-Acedo, and A. Nicolae. *Advances in Applied Mathematics*, 2025.

Maximum entropy on the mean and the Cramér rate function in statistical estimation and inverse problems: properties, models, and algorithms
With Y. Vaisbourd, R. Choksi, T. Hoheisel, and C.-B. Schönlieb. *Mathematical Programming*, 2025.

Power flow geometry and approximation
With J. Maack and D. Sigler. *IEEE Transactions on Power Systems*, 2025.

From perspective maps to epigraphical projections
With M. P. Friedlander and T. Hoheisel. *Mathematics of Operations Research*, 2022.

SUBMISSIONS TO REFEREED JOURNALS

A finite set with noncompact closed convex hull in a Hadamard space
Preprint, arXiv:2609.06455. Under review, 2026.

Minimal enclosing balls via geodesics
With A. S. Lewis. *Preprint*, arXiv:2603.15488. Under review, 2026.

CONFERENCE PAPERS

Riemannian optimization applied to AC optimal power flow
With J. Maack and D. Sigler. *IEEE Power & Energy Society General Meeting*, 2024.

CONFERENCE PRESENTATIONS AND TALKS

Minimum Enclosing Ball Algorithms in Geodesic Spaces — July 2027 (Upcoming)
International Symposium on Mathematical Programming, Amsterdam — Invited Talk

A Cyclic Projections Approach to Minimum Enclosing Balls — Jan. 2027 (Upcoming)
Joint Mathematics Meetings, Chicago — Invited Talk

Minimum Enclosing Ball Algorithms in Geodesic Spaces — Dec. 2026 (Upcoming)
Canadian Mathematical Society Winter Meeting, Montreal — Invited Talk

Subgradient Methods on Nonpositively Curved Geodesic Spaces — Nov. 2026
INFORMS Annual Meeting, San Francisco — Prize Talk

Minimum Enclosing Ball Algorithms in Geodesic Spaces — Nov. 2026
INFORMS Annual Meeting, San Francisco — Invited Talk

Convex Optimization in Geodesic Spaces — Oct. 2026
ORIE Young Researchers Workshop, Cornell University — Invited Talk

Convex Hulls in Geodesic Spaces of Nonpositive Curvature — Sept. 2026
Geometry Seminar, The Pennsylvania State University — Invited Talk

Minimum Enclosing Ball Algorithms in Geodesic Spaces — Aug. 2026
Institute for Foundations of Data Science Research Expo, University of Wisconsin–Madison — Poster

Subgradient Methods on Nonpositively Curved Geodesic Spaces — June 2026
SIAM Conference on Optimization, University of Edinburgh — Invited Talk

Subgradient Methods in Geodesic Spaces — Oct. 2025
ORIE Young Researchers Workshop, Cornell University — Poster

Incremental Minimization in Spaces of Nonpositive Curvature — July 2025
International Conference on Continuous Optimization, University of Southern California — Talk

Incremental Minimization in Spaces of Nonpositive Curvature — May 2025
Optimization over Nonlinear Model Spaces Seminar, University of Seville — Invited Talk

An Invitation to Hadamard Space — Nov. 2024
Applied Mathematics Student Colloquium, Cornell University — Talk

Incremental Minimization in Spaces of Nonpositive Curvature — Nov. 2024
Midwest Optimization Meeting, University of Waterloo — Poster

A Geometric Approach to Optimal Power Flow Aug. 2023
NSF Mathematical Sciences Graduate Internship Virtual Symposium Talk

The Maximum Entropy on the Mean Method for Linear Inverse Problems July 2022
International Conference on Continuous Optimization, Lehigh University Talk

Epigraphical Projections in Nonsmooth Optimization Aug. 2020
Undergraduate Research Conference, McGill University Talk

TEACHING EXPERIENCE

Ithaca High School Math Seminar Instructor 2025
Cornell University, Ithaca High School

· Designed and taught a seminar on mathematical optimization for advanced high school students

Teaching Assistant 2023 – 2024
Cornell University

· MATH 2930: Differential Equations for Engineers (Summer 2024)
· MATH 2940: Linear Algebra for Engineers (Spring 2024)
· MATH 1920: Multivariable Calculus for Engineers (Fall 2023)

Teaching Assistant 2019 – 2021
McGill University

· MATH 248: Honours Vector Calculus (Fall 2021)
· PHYS 131: Mechanics and Waves (Fall 2019, 2020, and 2021)

Tutor 2019 – 2021
McGill University

· MATH 140 (Calculus 1), 141 (Calculus 2), 248 (Honours Vector Calculus), 251 (Honours Algebra 2), PHYS 101 (Introductory Physics - Mechanics), 131 (Mechanics and Waves), COMP 330 (Theory of Computation), ECSE 202 (Introduction to Software Development)

SERVICE AND OUTREACH

Peer Mentor 2023 – 2027
Center for Applied Mathematics, Cornell University

· Mentoring applied math Ph.D. students in their transition to graduate school

Undergraduate Research Conference Moderator 2026
Summer Research Conference, University of California San Diego

· Moderated undergraduate research presentations

Science Outreach Volunteer 2026
Expanding Your Horizons, Cornell University

· Planned and facilitated math games at an annual outreach event for students in grades 7-10

Undergraduate Mentor 2026
Association for Women in Mathematics, Cornell University

· Mentored an undergraduate mathematics student

High School Research Mentor 2026
Ithaca High School Math Seminar, Cornell University

· Mentored advanced high school students on a research project in optimization

Directed Reading Program Mentor 2024 – 2025
Cornell University

· Mentored undergraduate students in projects on differential geometry, probability, and optimization

Conference Session Organizer

· *International Symposium on Mathematical Programming (ISMP 2027, Upcoming)*

Reviewer

· *Mathematical Programming, Mathematics of Operations Research, Journal of Optimization Theory and Applications, SIAM Journal on Optimization, SIAM Journal on Applied Algebra and Geometry, IEEE Transactions on Power Systems*

SKILLS AND INTERESTS

Computational Skills	Scientific programming, primarily in Julia and Python, with experience in C++, C, OCaml, Java, and MATLAB; proficient in using AI tools to develop proofs, implement algorithms, and run numerical experiments
Research Interests	Nonsmooth Optimization, Convex Analysis, Non-Euclidean Geometry, Optimization Algorithms, Data Science
Languages	English (Native), French (Intermediate)